

Patuakhali Science and Technology University

4th semester (L-2, S-II) Final Examination of B.Sc. in Engg. (CSE), July-Dec-2022, Session: 2020-21

Course Code: MAT-221, Course Title: Mathematics-IV

Marks-70, Time: 3 hours, Credit: 3.00

[Figure in the right margin indicates full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions.

1. a) Define Fourier series and derive its complex form. 04
- b) Define the half range Fourier series 02
- c) Expand the Fourier series for the following function 08

$$f(x) = \begin{cases} 0 & \text{when } -\pi < x < 0 \\ \frac{\pi x}{4} & \text{when } 0 < x < \pi \end{cases}$$

and hence prove that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$

2. a) Explain Fourier integral and Fourier transformation with its applications 05
- b) Find the Fourier integral of the function $f(x) = e^{-x}, x \geq 0$ 04
- c) Find the Fourier sine transformation of $F(x)$ where 05

$$F(x) = \begin{cases} x, & 0 < x < 1 \\ 2-x, & 1 < x < 2 \\ 0, & x > 2 \end{cases}$$

3. a) Define Laplace transform. 02
- b) State first translation or shifting property. 02
- c) Find the Laplace transform of $e^{2t}(\cos 3t - \sin 4t)$. 02
- d) Applying the Laplace transformation solve the following ordinary differential equation 08

$$\frac{d^3 Y}{dt^3} + \frac{d^2 Y}{dt^2} = e^t + t + 1, Y(0) = Y'(0) = Y''(0) = 0$$

4. a) Define Harmonic function. 02
- b) State and prove the necessary condition of the Cauchy-Riemann equations. 05
- c) Find the conjugate harmonic function of $u = y^3 - 3x^2y$. 05
- d) 02

Find the modulus and principal argument of $\left(\frac{1 + (\sqrt{3})i}{1 - (\sqrt{3})i} \right)^2$.

5. a) Show that $f(z) = 2x + iy^2$ is nowhere analytic. 02
- b) State Cauchy's Integral formula for n^{th} order derivative. If $f(z)$ is analytic for all points inside of C and connected a simple closed curve C . a is any point inside C . Then 05

$$f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z-a} dz.$$

- c) Evaluate: $\oint_C \frac{ze^{z^2}}{(z-1)^3} dz$, where C is the circle. 04

d) Show that $\left(\frac{a^n}{n!}\right)^2 = \frac{1}{2\pi i} \oint_C \frac{a^n e^{az}}{n! z^{n+1}} dz$. 03

6. a) Define residue. 02
 b) State and prove Cauchy's residue theorem. 06

c) Evaluate the contour integration $\int_0^{2\pi} \frac{d\theta}{5 + 3 \sin \theta}$ 06

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Answer any 5 of the following questions.

1. a) Write down the applications of Fourier series in the field of CSE. 02
- b) Define Fourier series for even and odd functions and also write it in the complex form. 04
- c) Find the Fourier series for the following function: 08

$$f(x) = \begin{cases} -1, & \text{when } -\pi < x < -\frac{\pi}{2} \\ 0, & \text{when } -\frac{\pi}{2} < x < \frac{\pi}{2} \\ 1, & \text{when } \frac{\pi}{2} < x < \pi \end{cases}$$

and hence prove that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$

2. a) Explain the Fourier integral and Fourier transform. 03
- b) 06

Find the Fourier integral of the function: $f(x) = \begin{cases} 0, & \text{when } x < 0 \\ \frac{1}{2}, & \text{when } x = 0 \\ e^x, & \text{when } x > 0 \end{cases}$

- c) Derive the finite Fourier sine transform of the function: $F(x) = 2x, 0 < x < 4$ 05

3. a) Illustrate the uses of Laplace transformation in various fields. 02

- b) State three properties of Laplace transformation. 02

- c) Evaluate: $L^{-1}\left[\frac{1}{s^2(s^2+4)}\right]$ by using the convolution theorem. 04

Applying the Laplace transformation solve the following ordinary differential equation 06

d) $\frac{d^2Y}{dt^2} + 9Y = \cos 2t, Y(0) = 1, Y\left(\frac{\pi}{2}\right) = -1$

4. a) Define analytic function. 02

- b) State and prove the necessary condition of the Cauchy-Riemann equations 06

- c) Find the conjugate harmonic function of $u = e^{-x}(x \sin y - y \cos y)$. 06

5. a) Define singular point. 02

- b) If $f(z)$ is analytic for all points inside of C and connected a simple closed curve C . a is 05

any point inside C . Then $f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z-a} dz$.

- c) Evaluate: 05

i) $\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz$, where C is the circle $|z| = 3$.

ii) $\oint_C \frac{ze^{iz}}{(z+1)^3} dz$, where C is the circle.

02

d) State Cauchy's inequality.

06

a) Let $f(z)$ be analytic inside and on a simple closed curve C except at a finite number of singular points $a_1, a_2, a_3, \dots, a_n$, then

$$\oint_C f(z) dz = 2\pi i [\operatorname{Res}(a_1) + \operatorname{Res}(a_2) + \dots + \operatorname{Res}(a_n)].$$

02

b) Find the residue of $f(z) = \frac{e^z}{(z - \pi i)^2}$.

06

c) Evaluate the contour integration $\int_0^{2\pi} \frac{d\theta}{a + b \cos \theta}$

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- 1 Show that the function $u = e^x(x \cos y - y \sin y)$ is a harmonic function and find the conjugate harmonic of u 7
- 2 Write down the application of Fourier series
- 3 Define Fourier series for even and odd function 2
- 4 Find the Fourier series of the function $f(x) = x^2, -\pi < x < \pi$ 2
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